Transition metals and their chemistry

Copper sulfate is blue, potassium manganate(VII) purple, and iron catalyses the Haber process. These properties all come from one feature of the transition metals: an incomplete set of d electrons. This topic explains complex ions, their shapes and colours, the many oxidation states of vanadium and chromium, the reactions of metal aqua ions, and how transition metals act as catalysts.

SPEC 17.1 What makes a transition metal

WHAT THIS SPEC POINT ASKS

Know that transition metals are d-block elements forming one or more stable ions with incompletely filled d-orbitals.

KEY IDEA

A **transition metal** is a **d-block element** that forms **at least one stable ion with an incomplete d subshell** (between 1 and 9 d electrons).

- **d-block element**: its highest-energy electron goes into a **d subshell**. For Period 4, that is scandium to zinc.
- Not every d-block element is a transition metal. The test is about the **ions**, not the atom.

Element	Only stable ion	d electrons	Transition metal?
Scandium	Sc^{3+} = [Ar]	$3d^0$ (empty)	**No**
Titanium to copper	e.g. Ti^{3+} $3d^1$, Fe^{2+} $3d^6$, Cu^{2+} $3d^9$	partly filled	**Yes**
Zinc	Zn^{2+} = [Ar]$3d^{10}$	$3d^{10}$ (full)	**No**

- Copper counts even though Cu^+ is $3d^{10}$, because **Cu^{2+}** ($3d^9$) has an incomplete d subshell. One such ion is enough.
- Typical transition-metal properties all come from that partly filled d subshell: **variable oxidation states**, **coloured** ions, **complex** formation and **catalytic** activity.

MARK-SCHEME WORDING

- Mercury: "Hg^+ is $[Xe]4f^{14}5d^{10}6s^1$ and Hg^{2+} is $[Xe]4f^{14}5d^{10}$"; "d-block element as the last electron goes into a (5)d orbital"; "not a transition element as Hg^+ and Hg^{2+} do not have incompletely filled d orbitals" (WCH15 Jun 2023).
- DO NOT ACCEPT an answer based only on the **atom's** configuration. Use "d orbitals" (plural) or "d subshell".
- Nickel "forms stable ions with partially filled d orbitals" (WCH15 Jan 2021).

Often asked as: is Zn / Sc / Hg a transition metal? (use the configurations of the ions); MCQs on the definition.

COMMON MISTAKE

Defining a transition metal by the atom ("it has d electrons") instead of by its **ions**.

Not explaining what "d-block" means when asked (WCH15 Jun 2023 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Applying the definition to **4d and 5d** metals (Ag^+ is $[Kr]4d^{10}$; Hg^{2+} is $[Xe]4f^{14}5d^{10}$), so neither forms a coloured simple ion (WCH15 Jan 2022; Jun 2023).

SPEC 17.2 Electronic configurations of Sc–Zn and their ions

WHAT THIS SPEC POINT ASKS

Deduce the electronic configurations of the atoms and ions of Sc to Zn from atomic number and charge.

Rules

1. **4s fills before 3d** (4s is lower in energy in the empty atom).
2. **Chromium and copper** are exceptions: Cr is **$[Ar]3d^5 4s^1$** and Cu is **$[Ar]3d^{10}4s^1$**. A half-full or full d subshell is especially stable.
3. When the atom forms an ion, **4s electrons are lost first**, before any 3d electrons.

	Atom	Common ions
Sc (21)	$[Ar]3d^1 4s^2$	Sc^{3+} $[Ar]$

	Atom	Common ions
Ti (22)	$[Ar]3d^2 4s^2$	Ti^{3+} $[Ar]3d^1$; Ti^{4+} $[Ar]$
V (23)	$[Ar]3d^3 4s^2$	V^{3+} $[Ar]3d^2$; V^{2+} $[Ar]3d^3$
Cr (24)	**$[Ar]3d^5 4s^1$**	Cr^{3+} $[Ar]3d^3$; Cr^{2+} $[Ar]3d^4$
Mn (25)	$[Ar]3d^5 4s^2$	Mn^{2+} $[Ar]3d^5$
Fe (26)	$[Ar]3d^6 4s^2$	Fe^{2+} $[Ar]3d^6$; Fe^{3+} $[Ar]3d^5$
Co (27)	$[Ar]3d^7 4s^2$	Co^{2+} $[Ar]3d^7$
Ni (28)	$[Ar]3d^8 4s^2$	Ni^{2+} $[Ar]3d^8$
Cu (29)	**$[Ar]3d^{10} 4s^1$**	Cu^{+} $[Ar]3d^{10}$; Cu^{2+} $[Ar]3d^9$
Zn (30)	$[Ar]3d^{10} 4s^2$	Zn^{2+} $[Ar]3d^{10}$

WORKED EXAMPLE

Worked example: Mn^{3+} and Fe^{3+}

Mn is 25: $[Ar]3d^5 4s^2$. Remove 3 electrons: the two 4s first, then one 3d → **Mn^{3+} = $[Ar]3d^4$**.

Fe is 26: $[Ar]3d^6 4s^2$. Remove 3: $4s^2$, then one 3d → **Fe^{3+} = $[Ar]3d^5$**.

Electrons-in-boxes

- Each box is one orbital. Fill each 3d box singly first with **parallel spins**, then pair up with **opposite** spins.
- Fe^{2+} ($3d^6$): five boxes, the first holding a pair (↑↓) and the other four single (↑). The 4s box is **empty**.

MARK-SCHEME WORDING

- "$[Ar]3d^5$" for Fe^{3+} and Mn^{2+}; "$[Ar]3d^6$" for Fe^{2+}; "$[Ar]3d^{10}$" for Zn^{2+} and Cu^{+}.
- Boxes: half-arrows accepted; "allow all arrows pointing downwards" in the 3d subshell, but two arrows in one orbital must point **opposite** ways (WCH15 Jan 2025).

Often asked as: MCQ on a configuration; complete the boxes for an atom and its ion; evidence from successive ionisation energies.

COMMON MISTAKE

Removing 3d electrons before 4s, or leaving 4s occupied in Fe^{2+} (WCH15 Jan 2025 examiner report). Writing

3d for silver instead of 4d (WCH15 Jan 2022 examiner report).

SPEC 17.3 Why transition metals have variable oxidation numbers

WHAT THIS SPEC POINT ASKS

Understand why transition metals show variable oxidation numbers.

The reason

- The **4s and 3d** subshells have **similar energies**.
- So after the 4s electrons are lost, the 3d electrons can be lost too without a huge jump in ionisation energy. Successive ionisation energies rise **gradually**, with no big jump until all 4s and 3d electrons are gone.
- Losing a different number of electrons gives a different ion. The energy needed is paid back by **lattice energy**, **hydration energy** or **covalent bonding**, so several oxidation states can be stable.
- The **maximum** oxidation state up to Mn equals the **total of 4s + 3d electrons**: V +5 ($3d^3 4s^2$), Cr +6, Mn +7.

Metal	Common oxidation states
Ti	+2, +3, **+4**
V	+2, +3, +4, **+5**
Cr	+2, **+3**, **+6**
Mn	**+2**, +4, +6, **+7**
Fe	**+2**, **+3**
Co	**+2**, +3
Ni	**+2**
Cu	+1, **+2**

Which state is more stable?

- **Fe^{3+} ($3d^5$)** is more stable than Fe^{2+} ($3d^6$). In $3d^6$, one orbital holds a pair, and the extra repulsion makes that sixth electron easy to remove. So Fe^{2+} is readily oxidised by air.
- **Mn^{2+} ($3d^5$)** is more stable than Mn^{3+} ($3d^4$). Removing an electron would break the stable half-filled subshell. So Mn^{2+} is hard to oxidise, and the Mn^{3+}/Mn^{2+} E° is very positive.
- High oxidation states exist only in **oxo-ions or covalent compounds**, never as bare ions. A V^{5+} ion would be tiny and highly charged: it would polarise the surrounding water molecules so strongly that they lose H^+, giving

VO_2^+.

MARK-SCHEME WORDING

- Highest state of V: "+5"; "vanadium has 3 electrons in the 3d and 2 in the 4s"; "the total of 3d and 4s electrons gives the highest oxidation state" (WCH15 Oct 2022).
- Mn^{2+} v Fe^{2+}: "there is stability associated with a half-full set of (3)d orbitals"; "Mn^{2+} has five d electrons so is more stable than Mn^{3+}"; "Fe^{2+} has six d electrons so is less stable than Fe^{3+}" (WCH15 Jan 2023); "$3d^6$ has two paired electrons which results in repulsion" (WCH15 Jun 2021).
- V^{5+}: "small / highly charged"; "so it would polarise water molecules / O–H bonds"; "causing them to lose hydrogen ions" (WCH15 Jan 2023).

Often asked as: why Fe^{2+} solutions go brown in air but Mn^{2+} solutions don't; the highest oxidation state from a configuration; why V^{5+} doesn't exist in water.

COMMON MISTAKE

Writing [Ar]$3d^3$ for vanadium (forgetting 4s) and deducing +3 as the maximum (WCH15 Oct 2022 examiner report).

Comparing two **elements** when the question is about two **ions** of the same element. Mixing up orbital, subshell and shell (WCH15 Jan 2023; Jan 2025 examiner reports).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The $Fe^{2+} \rightarrow Fe^{3+}$ argument with energies: the third ionisation energy ($Fe^{2+}(g) \rightarrow Fe^{3+}(g) + e^-$) is large, but it is **recovered by the more exothermic hydration** of Fe^{3+}, which has the greater charge (charge density) (WCH15 Jun 2021).

SPEC 17.4 Ligands

WHAT THIS SPEC POINT ASKS

Know what a ligand is.

KEY IDEA

A **ligand** is a molecule or ion with **at least one lone pair of electrons** that it donates to a central metal ion to form a **dative (coordinate) bond**.

- Ligands are **Lewis bases** (electron-pair donors).
- Common ones: H_2O and NH_3 (lone pair on O and N), Cl^-, OH^-, CN^-.
- A species **without a lone pair** can't be a ligand. NH_4^+ has none left (all are used in N–H bonds).

MARK-SCHEME WORDING

- "(a species with a) lone pair (of electrons) that can form a dative (covalent) bond to a (central transition) metal (ion)". ACCEPT "coordinate" for dative (WCH15 Oct 2021).
- MCQ answer: a ligand is an "electron-pair donor" (WCH15 Jun 2021).

Often asked as: define a ligand / a monodentate ligand (2 marks); why a species can't act as a ligand.

COMMON MISTAKE

Leaving out the words "lone pair". Saying a monodentate ligand "contains only one lone pair": H_2O has two lone pairs but uses **one** (WCH15 Oct 2021 examiner report).

SPEC 17.5 Dative covalent bonding in complex ions

WHAT THIS SPEC POINT ASKS

Understand that dative (coordinate) bonds form when complex ions are made.

- In a **dative covalent bond**, **both** electrons of the shared pair come from the **same** atom: the ligand's donor atom.
- In $[Cu(H_2O)_6]^{2+}$, each water uses **one lone pair on its oxygen atom** to bond to Cu^{2+}. The metal ion has empty orbitals that accept the pairs.
- Once formed, a dative bond is the same as any other covalent bond. In diagrams, draw it as an arrow **from the donor atom to the metal** (→) or as a plain line.
- A complex ion therefore contains **covalent** bonds (inside the ligands) and **dative covalent** bonds (ligand to metal), but no ionic bonds inside the brackets.

MARK-SCHEME WORDING

- "forms one dative (covalent) / coordinate bond (to Fe^{2+}) using a lone pair (of electrons) on **oxygen**" (WCH15 Oct 2023).
- Bonding in a complex: covalent and dative covalent only (WCH15 Jan 2022).

COMMON MISTAKE

Saying the lone pair comes from "the water molecule" instead of **the oxygen atom** (WCH15 Oct 2023 examiner report).

Drawing a dative arrow from H in NH_3: the bond comes from **N** (WCH15 Jun 2025).

SPEC 17.6 Complex ions

WHAT THIS SPEC POINT ASKS

Know that a complex ion is a central metal ion surrounded by ligands.

- A **complex** is a central metal atom or ion with ligands bonded to it by dative bonds. If it carries a charge, it is a **complex ion**.
- The whole complex is written in **square brackets**, with the overall charge outside.
- **Overall charge = charge on the metal ion + total charge of the ligands.**

Complex	Metal ion	Ligands	Overall charge
$[Cu(H_2O)_6]^{2+}$	Cu^{2+}	6 H_2O (neutral)	2+
$[CuCl_4]^{2-}$	Cu^{2+}	4 Cl^-	+2 − 4 = 2−
$[Fe(CN)_6]^{3-}$	Fe^{3+}	6 CN^-	+3 − 6 = 3−
$[Cr(OH)_6]^{3-}$	Cr^{3+}	6 OH^-	3−
$[Pt(NH_3)_2Cl_2]$	Pt^{2+}	2 NH_3, 2 Cl^-	0 (neutral)

WORKED EXAMPLE

Worked example: working back to the metal's charge

$[Ni(C_2O_4)_2Cl_2]^x$: Ni^{2+}, two $C_2O_4^{2-}$ (−4) and two Cl^- (−2) → x = +2 − 4 − 2 = **4−**.

$[Co(NH_3)_5Cl]^{2+}$: NH_3 neutral, Cl^- is −1, so Co − 1 = +2 → **Co is +3**.

MARK-SCHEME WORDING

- Formula of the complex in $FeCl_2{\cdot}4H_2O$ dissolved in water: $[Fe(H_2O)_6]^{2+}$ (ACCEPT $[FeCl(H_2O)_5]^+$) (WCH15 Oct 2021).
- Overall charge and coordination number deduced from the ligands (WCH15 Oct 2020; Jun 2024).

Often asked as: deduce a charge; write the formula of the complex ion in a solution.

COMMON MISTAKE

Giving just "Fe^{2+}", a neutral complex, or $Fe(OH)_2$ when asked for the complex ion in solution (WCH15 Oct 2021 examiner report).

Forgetting the charge on an anionic ligand such as SCN^- when writing the formula (WCH15 Oct 2024 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Mass of chlorine **in the complex ion only**, not in the whole compound (e.g. in $[Co(NH_3)_5Cl]Cl_2$ only one Cl is a ligand) (WCH15 Oct 2024).

SPEC 17.7 Aqueous transition-metal ions are usually coloured

WHAT THIS SPEC POINT ASKS

Know that aqueous solutions of transition-metal ions are usually coloured.

Ion (as hexaaqua ion unless shown)	d electrons	Colour
Sc^{3+}	$3d^0$	colourless
Ti^{3+}	$3d^1$	violet
VO_2^+ (+5) / VO^{2+} (+4) / V^{3+} / V^{2+}	$3d^0$ / $3d^1$ / $3d^2$ / $3d^3$	yellow / blue / green / violet
Cr^{3+}	$3d^3$	green (violet when pure hexaaqua)
$Cr_2O_7^{2-}$ / CrO_4^{2-} (+6)	$3d^0$	orange / yellow
Mn^{2+}	$3d^5$	very pale pink (looks colourless)
MnO_4^- (+7)	$3d^0$	purple
Fe^{2+}	$3d^6$	pale green

Ion (as hexaaqua ion unless shown)	d electrons	Colour
Fe^{3+}	$3d^5$	yellow / orange-brown
Co^{2+}	$3d^7$	pink
Ni^{2+}	$3d^8$	green
Cu^{2+}	$3d^9$	blue
Cu^{+} (in complexes)	$3d^{10}$	colourless
Zn^{2+}	$3d^{10}$	colourless

The colours of d^0 oxo-ions (MnO_4^-, $Cr_2O_7^{2-}$) come from a different process, charge transfer, and need no explanation at this level.

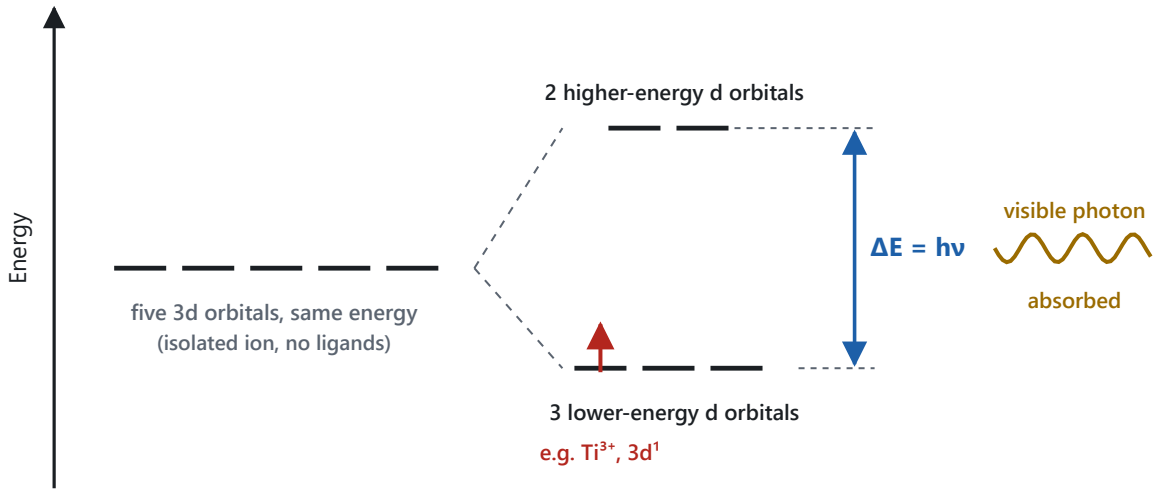
EXAM TIP

Learn these as a block: they feed straight into the NaOH/NH_3 table (17.22) and the vanadium and chromium sections. Every "describe what you see" question needs the **starting colour**.

SPEC 17.8 Why complex ions are coloured: d-orbital splitting

WHAT THIS SPEC POINT ASKS

Understand that colour arises because ligands split the energy levels of the d orbitals.



Ligands split the d orbitals. An electron in the lower set absorbs a photon of energy ΔE and is promoted.

In an octahedral complex the five 3d orbitals split into a lower set of three and a higher set of two.

The explanation, step by step

1. In an isolated ion, the five 3d orbitals all have the **same energy**.
2. When **ligands** bond to the ion, their lone pairs repel the d electrons unequally, so the d orbitals **split** into two groups with an **energy gap, ΔE**.
3. An electron in the lower group can absorb a photon of **visible light** whose energy exactly matches ΔE, and is **promoted** to the higher group (a **d–d transition**).
4. The light of that frequency is **absorbed**. The rest is **transmitted** (or reflected), and that is the colour you see: the **complementary colour** of the light absorbed.

$\Delta E = h\nu = hc/\lambda$, where $h = 6.63 \times 10^{-34}$ J s, ν = frequency (Hz), $c = 3.00 \times 10^{8}$ m s^{-1}, λ = wavelength (m).

- A **bigger** ΔE means **higher frequency** (shorter wavelength) light is absorbed.

Light absorbed	Colour seen
red	green / blue-green
orange	blue
yellow	violet
green	purple / magenta
blue	orange
violet	yellow

WORKED EXAMPLE

Worked example: ΔE = hν

$[Ti(H_2O)_6]^{3+}$ absorbs most strongly at λ = 500 nm (green-yellow light).

$\nu = c/\lambda = 3.00 \times 10^{8} \div 500 \times 10^{-9} =$ **6.00×10^{14} Hz**

$\Delta E = h\nu = 6.63 \times 10^{-34} \times 6.00 \times 10^{14} =$ **3.98×10^{-19} J** per ion

Per mole: $3.98 \times 10^{-19} \times 6.02 \times 10^{23} = 2.39 \times 10^{5}$ J mol^{-1} = **239 kJ mol^{-1}**

Green-yellow light is absorbed, so the solution looks **violet**.

For comparison, $[Cu(H_2O)_6]^{2+}$ absorbs around 800 nm (red/orange): $\Delta E = 2.49 \times 10^{-19}$ J, a **smaller** gap, and the solution looks blue.

MARK-SCHEME WORDING

The chain the marks follow (WCH15 Jun 2021; Oct 2024; Jan 2026):

- "ligands **split** the energy of the **d subshell / d orbitals**";
- "(visible) light is **absorbed** and promotes electrons from lower to higher energy (d orbitals)";
- "colour seen is the light **transmitted / reflected** / the wavelengths not absorbed / the complementary colour".
- Different colours: "different ligands / oxidation states → different **energy gap** → different wavelength / frequency absorbed".
- Always "d orbitals" (plural) or "d subshell". DO NOT ACCEPT "d orbital" (singular); penalise "shell" for subshell once.

Often asked as: explain why $[M(H_2O)_6]^{n+}$ is coloured (3 marks); 6-mark on why two complexes of one metal differ in colour; why an indicator-free titration has a visible end point.

COMMON MISTAKE

"Light is **emitted**" (that's a flame test): DO NOT ACCEPT. "**Adsorbed**" for absorbed. "A d orbital splits" (WCH15 Oct 2021; Jan 2024; Oct 2024 examiner reports).

Vague "different energy levels" instead of "a different **energy gap**" (WCH15 Oct 2024 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Colorimetry: the absorbance of a coloured complex is proportional to its concentration, so a **calibration graph** of absorbance against known concentration gives an unknown concentration. Use a filter of the **complementary** colour (WCH15 Jun 2021; Oct 2024).

f-block ions: lanthanide colours come from f–f transitions; La^{3+} (no f electrons) is colourless (WCH15 Oct 2020; Oct 2025).

SPEC 17.9 Why some ions are colourless

WHAT THIS SPEC POINT ASKS

Understand why some aqueous ions and complex ions have no colour.

A d–d transition needs an electron in the lower set **and** a space in the upper set. So there's no colour when:

Case	Example	Reason
d^0: empty d subshell	Sc^{3+}, Ti^{4+}	no d electrons to promote
d^{10}: full d subshell	Zn^{2+}, Cu^+, Ag^+	upper set already full, so no transition possible
ΔE outside the visible range	some complexes	absorption is in the UV, so no visible light is absorbed

MARK-SCHEME WORDING

- Cu(I): "the d orbitals are **full**, so no (electron) transitions are possible / no d–d transitions" (WCH15 Jan 2021).
- Cr(VI): "chromium(VI) has an **empty** d subshell" (WCH15 Jun 2022).
- DO NOT ACCEPT "the subshell is not split", "the light is outside the visible region" or "no electrons in d orbitals" **for Cu(I)**.

Often asked as: explain why a Cu(I) or Ag(I) complex is colourless; why Zn^{2+} solutions are colourless.

COMMON MISTAKE

Saying a d^{10} ion has "no d electrons". It has ten: the point is that there's **nowhere to promote** them.

SPEC 17.10 Coordination number

WHAT THIS SPEC POINT ASKS

Understand what coordination number means.

KEY IDEA

The **coordination number** is the **number of dative bonds** from ligands to the central metal ion.

- For monodentate ligands, it equals the number of ligands. For multidentate ligands, it **doesn't**: $[Cu(en)_3]^{2+}$ has 3 ligands but coordination number **6** (each en forms 2 bonds).

Coordination number	Shape	Typical ligands	Example
6	octahedral	small ligands: H_2O, NH_3, OH^-	$[Fe(H_2O)_6]^{2+}$
4	tetrahedral	large ligands, e.g. Cl^-	$[CuCl_4]^{2-}$
4	square planar	Pt^{2+} and Ni^{2+} complexes	$[Pt(NH_3)_2Cl_2]$
2	linear	Ag^+	$[Ag(NH_3)_2]^+$

MARK-SCHEME WORDING

- "6 and octahedral" for three bidentate ligands; coordination number is **not** the number of ligands (WCH15 Jun 2021; Jun 2024).
- Deducing a formula from data: "a 1 : 1 ratio indicates one thiocyanate ion in an octahedral complex, so 5 water molecules": $[Fe(H_2O)_5SCN]^{2+}$ (WCH15 Oct 2024).

COMMON MISTAKE

Giving 3 for $[M(en)_3]^{n+}$. Count **bonds**, not ligands (WCH15 Jun 2024 examiner report).

SPEC 17.11 Why colours change: oxidation number, ligand and coordination number

WHAT THIS SPEC POINT ASKS

Understand that the colour of a transition-metal ion can change because of a change in (i) oxidation number, (ii) ligand, (iii) coordination number.

Each change alters the **size of ΔE**, so a different frequency of light is absorbed and a different colour is seen.

Change	Example	Colour change
Oxidation number	$[Fe(H_2O)_6]^{2+} \rightarrow [Fe(H_2O)_6]^{3+}$	pale green → yellow-brown
	$VO_2^+ \rightarrow VO^{2+} \rightarrow V^{3+} \rightarrow V^{2+}$	yellow → blue → green → violet
Ligand (same coordination number)	$[Cu(H_2O)_6]^{2+} \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}$	blue → deep blue
	$[Co(H_2O)_6]^{2+} \rightarrow [Co(NH_3)_6]^{2+}$	pink → straw / yellow-brown
Coordination number (and ligand)	$[Cu(H_2O)_6]^{2+}$ (6) → $[CuCl_4]^{2-}$ (4)	blue → yellow (green in between)
	$[Co(H_2O)_6]^{2+}$ (6) → $[CoCl_4]^{2-}$ (4)	pink → blue

MARK-SCHEME WORDING

- "the ammonia ligands split the two groups of orbitals by a different amount than water"; "the cobalt is in a different oxidation state, which splits … by a different amount"; "so different wavelengths / frequencies of light are absorbed" (WCH15 Jan 2026).
- ALLOW "a different number of electrons in the d orbitals" for the oxidation-state effect.

Often asked as: MCQ colour sequences; 6-mark on why two cobalt complexes have different colours.

COMMON MISTAKE

Saying the colour changes "because the ligand is different" and stopping. Link it to a **different ΔE** and **different frequency absorbed**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

$[Co(NH_3)_6]^{2+}$ is slowly **oxidised by air** to $[Co(NH_3)_6]^{3+}$: the straw solution darkens to brown. A change of oxidation state with the same ligands (WCH15 Jan 2026).

SPEC 17.12 H_2O, OH^- and NH_3 as monodentate ligands

WHAT THIS SPEC POINT ASKS

Understand that H_2O, OH^- and NH_3 act as monodentate ligands.

- A **monodentate** ligand forms **one** dative bond to the metal ion, through **one** donor atom.
- **H_2O**: lone pair on **O** (it has two lone pairs but can only use one at a time).
- **OH^-**: lone pair on **O**; it carries a negative charge, so it lowers the complex's overall charge.
- **NH_3**: the lone pair on **N**.
- H_2O and NH_3 are similar in size and both neutral, so swapping one for the other doesn't change the coordination number. Cl^- is also monodentate, but bigger.

MARK-SCHEME WORDING

- "monodentate: forms a single / one dative (covalent) bond"; ACCEPT "occupies one coordination site" (WCH15 Oct 2021).

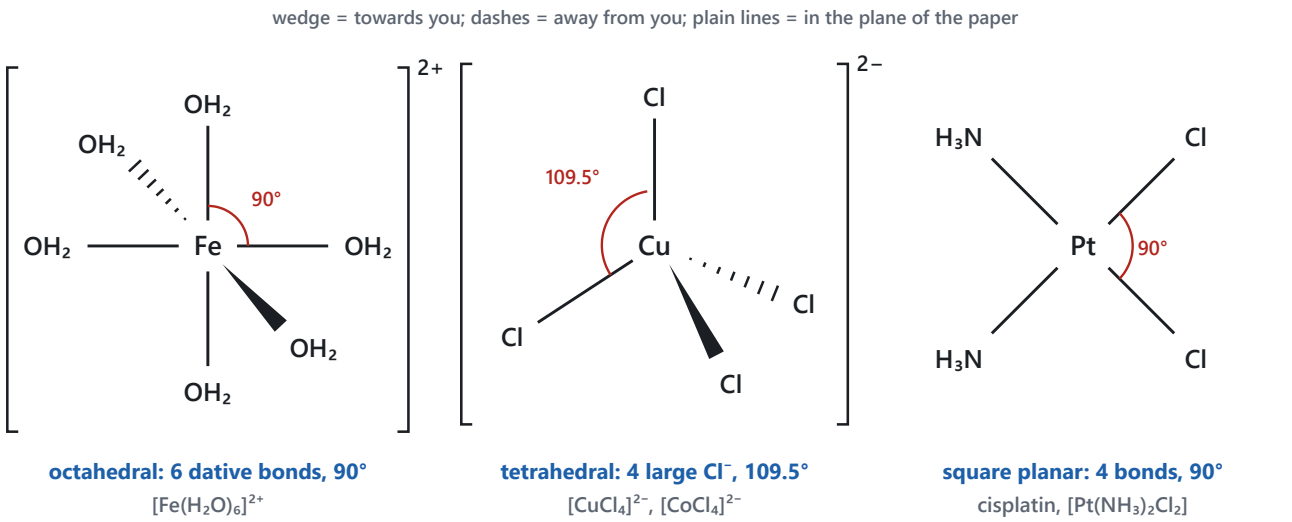
COMMON MISTAKE

"Monodentate means it has one lone pair." It means it forms **one** bond.

SPEC 17.13 Octahedral complexes

WHAT THIS SPEC POINT ASKS

Understand why six-fold coordination gives an octahedral shape, as in complexes with H_2O, OH^- and NH_3.



The three shapes you need, with bond angles. Draw wedges and dashes for 3D shapes.

- Six ligands mean **six bonding pairs** around the metal ion.
- Electron pairs **repel**, so they get **as far apart as possible**: an **octahedron**, all bond angles **90°** (180° between opposite ligands).
- Small ligands (H_2O, NH_3, OH^-) fit six around a Period 4 ion: $[Fe(H_2O)_6]^{2+}$, $[Ni(NH_3)_6]^{2+}$, $[Cr(OH)_6]^{3-}$.
- Three bidentate ligands or one hexadentate ligand also give an octahedral complex.

Drawing it

Four bonds in the plane of the paper (two vertical, two horizontal, plain lines), one **wedge** (towards you) and one **dashed** bond (away). Show the metal symbol, all six donor atoms bonded through the right atom (O of H_2O, N of NH_3), square brackets and the overall charge.

MARK-SCHEME WORDING

- "**six pairs of (bonding) electrons**, which are as far apart as possible to **minimise repulsion**" (WCH15 Oct 2023).
- 3D diagrams: wedge and dash bonds, 90°, correct charge. "Penalise lack of / incorrect wedges once only"; dative arrows accepted (WCH15 Oct 2025).

Often asked as: draw a 3D diagram of an octahedral complex with bond angles; explain the shape.

COMMON MISTAKE

Explaining the shape by "six ligands" alone, instead of "six **pairs of electrons** repelling" (WCH15 Oct 2023 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

VSEPR for other complexes: $[Ag(NH_3)_2]^+$ is **linear, 180°**, because it has two bonding pairs that minimise repulsion (WCH15 Jan 2022).

Isomerism in octahedral complexes: cis/trans isomers (e.g. two Cl^- next to or opposite each other in $[Co(NH_3)_4Cl_2]^+$) and **optical isomers** with three bidentate ligands ($[Ni(en)_3]^{2+}$): two non-superimposable mirror images (WCH15 Jan 2024).

SPEC 17.14 Tetrahedral complexes with large ligands such as Cl^-

WHAT THIS SPEC POINT ASKS

Know that transition-metal ions may form tetrahedral complexes with relatively large ions such as Cl^-.

- Chloride ions are **much larger** than H_2O or NH_3. Only **four** fit around a Period 4 metal ion.
- Four bonding pairs repel to be as far apart as possible: **tetrahedral**, **109.5°**.
- Examples: **$[CuCl_4]^{2-}$** (yellow), **$[CoCl_4]^{2-}$** (blue). Both are made by adding **concentrated HCl** to the aqua ion (17.24).

MARK-SCHEME WORDING

- "**chloride ions are large**, so steric hindrance is too great for six ligands to fit around the central ion". ACCEPT "chloride ions are large". DO NOT ACCEPT "chlorine atoms". IGNORE "repulsion between negative chloride ions" (WCH15 Jun 2021).
- "$[CoCl_4]^{2-}$ is tetrahedral because Cl^- are too large for six to coordinate around Co^{2+}" (WCH15 Oct 2022).

Often asked as: why the chloro complex is tetrahedral (1–2 marks); draw $[CuCl_4]^{2-}$ in 3D with 109.5°.

COMMON MISTAKE

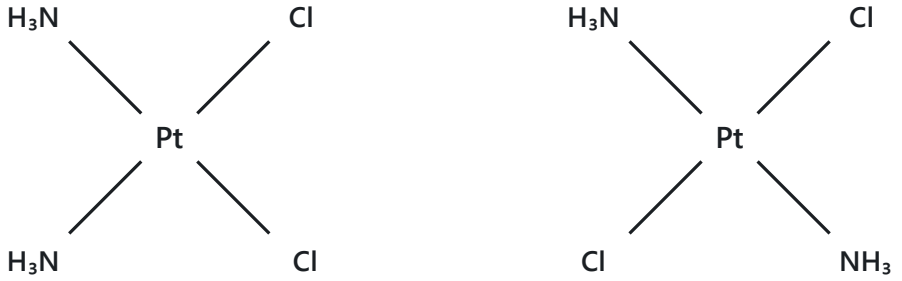
Writing "chlorine" or "chlorine atoms" for the chloride **ion** (WCH15 Oct 2022 examiner report).

SPEC 17.15 Square planar complexes and cisplatin

WHAT THIS SPEC POINT ASKS

Know that square planar complexes exist and that cisplatin is one, used in cancer treatment as a single isomer (not mixed with the trans form).

- Some four-coordinate complexes are **square planar**: all four bonds in one plane, **90°** apart. Pt^{2+} and Ni^{2+} form them, e.g. $[Pt(NH_3)_2Cl_2]$, $[Ni(CN)_4]^{2-}$.
- **Cisplatin**, $[Pt(NH_3)_2Cl_2]$, is a neutral platinum(II) complex. The two Cl^- ligands are **next to each other** (cis, 90° apart).
- In **transplatin** the two Cl^- are **opposite** each other (180°). Same formula, different arrangement: **cis/trans (geometric) isomers**.



cis-platin (the drug)
both Cl^- on the same side

trans-platin (not used)
Cl^- opposite each other

Only the cis isomer is an anticancer drug.

How cisplatin works

1. Inside a cell, $[Cl^-]$ is low and $[H_2O]$ is very high, so a **Cl^- ligand is replaced by water** (ligand exchange): $[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$.
2. The water ligand is then displaced by a **nitrogen atom on a guanine base** in DNA, which donates its lone pair to Pt. The second Cl^- is replaced the same way by a second, **neighbouring** guanine.
3. Pt now cross-links two adjacent bases on the same strand, so the DNA **kinks** and can't be **copied**. The cancer cell can't divide and dies.

- **Why only cis**: the two leaving groups must be on the **same side** to bind two neighbouring bases. In transplatin the second Cl^- is on the opposite side, too far from the DNA strand.
- It is given as **the pure cis isomer**, because the trans form doesn't work and would add side effects without any benefit.
- **Side effects**: cisplatin also attacks healthy, rapidly dividing cells (hair follicles, gut lining), and it damages the kidneys. Doses are balanced against these effects.

MARK-SCHEME WORDING

- "square planar"; bonds drawn from **N** (not H) to Pt (WCH15 Jun 2025).
- "there is a higher concentration of water (than chloride ions) in cells, so the equilibrium moves to the right" (Cl^- replaced by H_2O) (WCH15 Jun 2025).
- Equation "$[Pt(NH_3)_2Cl_2] + H_2O \rightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$" ("do not award products without charges"); "a lone pair on nitrogen is donated to platinum"; trans: "the second chloride is too far from / on the opposite side to the DNA strand" (WCH15 Jan 2024).

Often asked as: shape of cisplatin; 4–7 mark question on how it binds DNA and why the trans isomer is not used.

COMMON MISTAKE

Missing the **+ charge** on $[Pt(NH_3)_2(H_2O)Cl]^+$, or losing an NH_3 ligand instead of Cl^- (WCH15 Jan 2024 examiner report).

Being vague about **which side** the DNA is on for transplatin.

SPEC 17.16 Bidentate and hexadentate ligands

WHAT THIS SPEC POINT ASKS

Understand "bidentate" and "hexadentate" and identify examples such as $NH_2CH_2CH_2NH_2$ and $EDTA^{4-}$.

Ligand	Donor atoms	Dative bonds per ligand	Example complex
Ethane-1,2-diamine, "en", $H_2NCH_2CH_2NH_2$	2 N (one lone pair on each)	**2** (bidentate)	$[Cu(en)_3]^{2+}$, octahedral, CN 6
Ethanedioate, $C_2O_4^{2-}$	2 O (one on each carboxylate)	**2** (bidentate)	$[Fe(C_2O_4)_3]^{3-}$
EDTA^{4-}	2 N + 4 O (carboxylate groups)	**6** (hexadentate)	$[Cu(EDTA)]^{2-}$, octahedral
Haem (porphyrin)	4 N	**4** (polydentate)	haemoglobin (17.17)

- **Bidentate**: two donor atoms, two dative bonds. **Hexadentate**: six donor atoms, six dative bonds. **Polydentate / multidentate**: more than one.

- These ligands wrap round the metal ion forming rings, so they are also called **chelating** ligands ("chela" = claw). en forms a stable **five-membered ring** M–N–C–C–N.
- The donor atoms must be **far enough apart** along the chain to reach two sites next to each other on the metal ion.

MARK-SCHEME WORDING

- $EDTA^{4-}$: "a **hexadentate** ligand / can form 6 dative (covalent) bonds"; "complex is more stable / is a chelating agent / traps the Cr^{3+}" (WCH15 Jan 2025).
- en diagram: "6 bonds between N in the diamines and Fe". DO NOT ACCEPT the two N of one en bonded at 180° (WCH15 Oct 2020).
- Use the spec terms: DO NOT ACCEPT "didentate", or "polydentate" when "bidentate" is the answer (WCH15 Jun 2024; Jun 2025).

Often asked as: identify a ligand as bidentate/hexadentate from a structure; draw $[M(en)_3]^{n+}$; why $EDTA^{4-}$ treats metal poisoning.

COMMON MISTAKE

Assuming an unfamiliar ligand is bidentate without counting its donor atoms (salen has 2 O and 2 N, so it is tetradentate) (WCH15 Oct 2022 examiner report).

Giving coordination number 3 for three en ligands.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Unfamiliar polydentate ligands in context: salen (2 O, 2 N), thioglycolate (S and O), carminic acid, diphenylcarbazide (WCH15 Oct 2022; Jun 2021; Oct 2025; Jun 2022).

Chelation therapy: $EDTA^{4-}$ binds toxic metal ions (e.g. Cr^{3+}, Pb^{2+}) so they can be excreted (WCH15 Jan 2025).

SPEC 17.17 Haemoglobin and carbon monoxide

WHAT THIS SPEC POINT ASKS

Know that haemoglobin is an iron(II) complex with a polydentate ligand, and that ligand exchange happens when CO replaces O_2 bound to haemoglobin.

- **Haem** contains an **Fe^{2+}** ion held by a **polydentate** ring ligand that bonds through **four nitrogen atoms**, all in one plane.
- A fifth dative bond comes from an N in the **globin** protein; the **sixth** site binds **O_2** (or H_2O). Six bonds: **octahedral**.
- O_2 binds **reversibly** and fairly weakly, so it is picked up in the lungs and released in respiring tissues.
- **Carbon monoxide** is a better ligand: it binds to Fe^{2+} about **200 times more strongly**. It **replaces** O_2 by **ligand exchange**:

$HbO_2 + CO \rightleftharpoons HbCO + O_2$

- The equilibrium lies far to the right, so CO is hard to remove: haemoglobin that has bound CO **can't carry oxygen**. Symptoms of poisoning follow from oxygen starvation of tissues.
- (The structure of haem is not assessed.)

MARK-SCHEME WORDING

- "carbon monoxide **replaces / takes the place of the oxygen** molecule / ligand"; "because it **binds more strongly** to the Fe^{2+} ion". ACCEPT "forms a more stable complex", "larger equilibrium constant", "binds (almost) irreversibly" (WCH15 Oct 2023).
- "Prevents oxygen being carried" scores only the second mark: the replacement of O_2 must be stated.

Often asked as: why CO is toxic (2 marks); MCQ on how O_2 bonds to Fe^{2+}.

COMMON MISTAKE

Only saying CO "stops oxygen being carried" without saying it **replaces O_2** by **ligand exchange** (WCH15 Oct 2023 examiner report).

SPEC 17.18 Colours of vanadium's oxidation states

WHAT THIS SPEC POINT ASKS

Know the colours of vanadium in oxidation states +5, +4, +3 and +2.

Oxidation state	Species in acid solution	Name	Colour
+5	**VO_2^+** (VO_3^- in the solid / alkali)	dioxovanadium(V)	**yellow**
+4	**VO^{2+}**	oxovanadium(IV)	**blue**
+3	**V^{3+}**	vanadium(III)	**green**

Oxidation state	Species in acid solution	Name	Colour
+2	V^{2+}	vanadium(II)	**violet** (lilac)

Memory aid: "**Y**ou **B**etter **G**et **V**anadium" (yellow, blue, green, violet as the number goes down).

- When zinc reduces yellow VO_2^+, a **green** colour appears before the blue: that's a **mixture** of yellow V(+5) and blue V(+4), not V^{3+}. So the full sequence you see is **yellow** → **green** → **blue** → **green** → **violet**.

MARK-SCHEME WORDING

- "+3 green; +4 blue; +5 yellow" (WCH15 Oct 2020). DO NOT ACCEPT combined colours such as "blue/green".
- The +5 species is **VO_2^+** and the +4 species is **VO^{2+}**, not V^{5+} or V^{4+} (WCH15 Jan 2023).

Often asked as: colours of each oxidation state; MCQ on which colour is +3.

COMMON MISTAKE

Mixing up VO_2^+ (V +5) and VO^{2+} (V +4). Check with oxidation numbers: in VO_2^+, V − 4 = +1, so V = +5.

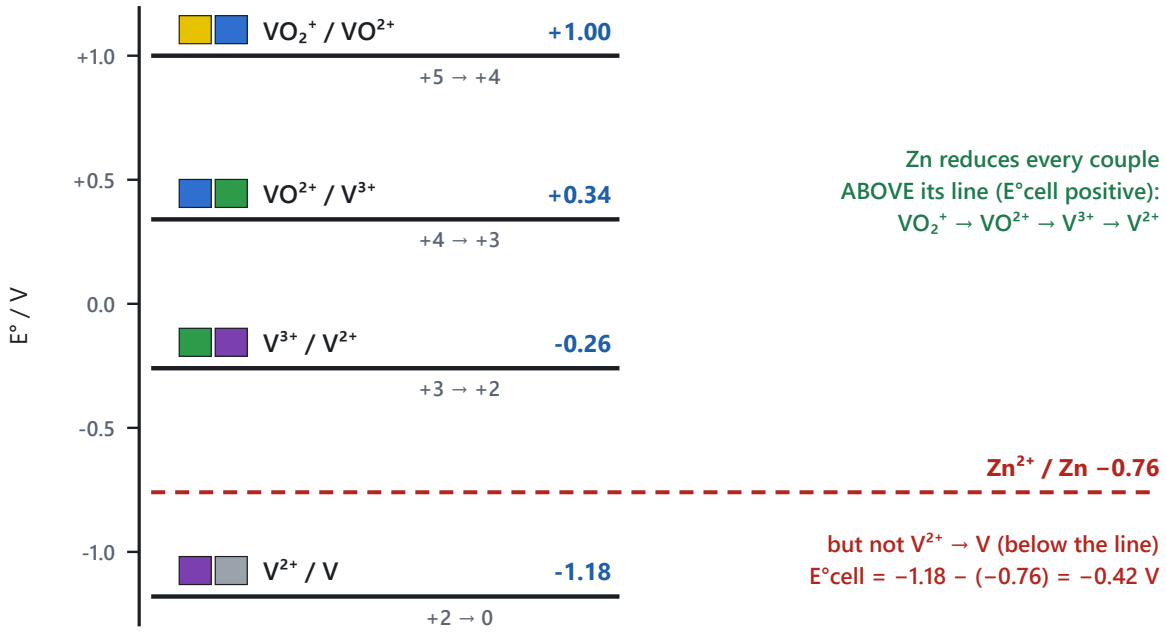
Writing V^{5+} as if it existed as a simple ion (WCH15 Jan 2023 examiner report).

SPEC 17.19 Interconverting vanadium's oxidation states using E°

WHAT THIS SPEC POINT ASKS

Understand the redox reactions that interconvert vanadium's +5, +4, +3 and +2 states, using the relevant E° values.

Half-equation	E° / V
$VO_2^+ + 2H^+ + e^- \rightleftharpoons VO^{2+} + H_2O$	+1.00
$VO^{2+} + 2H^+ + e^- \rightleftharpoons V^{3+} + H_2O$	+0.34
$V^{3+} + e^- \rightleftharpoons V^{2+}$	−0.26
$V^{2+} + 2e^- \rightleftharpoons V$	−1.18



Colours: yellow (+5), blue (+4), green (+3), violet (+2)

Zinc (−0.76 V) is more negative than each of the top three vanadium couples, so it reduces V(+5) all the way to V(+2), but not to the metal.

Reduction with zinc in acid

Shake ammonium vanadate(V), NH_4VO_3, with **zinc and dilute acid**; put a loose **cotton-wool plug** in the flask so hydrogen can escape but air can't get in (air would oxidise V^{2+} back).

WORKED EXAMPLE

Worked example: zinc reduces vanadium step by step

$Zn^{2+} + 2e^- \rightleftharpoons Zn$, E° = −0.76 V

+5 → +4: E°cell = 1.00 − (−0.76) = +1.76 V ✓

$2VO_2^+ + 4H^+ + Zn \rightarrow 2VO^{2+} + Zn^{2+} + 2H_2O$

+4 → +3: E°cell = 0.34 − (−0.76) = +1.10 V ✓

$2VO^{2+} + 4H^+ + Zn \rightarrow 2V^{3+} + Zn^{2+} + 2H_2O$

+3 → +2: E°cell = −0.26 − (−0.76) = +0.50 V ✓

$2V^{3+} + Zn \rightarrow 2V^{2+} + Zn^{2+}$

+2 → 0: E°cell = −1.18 − (−0.76) = −0.42 V ✗ **not feasible**, so the reduction stops at **V^{2+}**.

WORKED EXAMPLE

Worked example: a weaker reducing agent stops sooner

Fe^{2+} ($Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$, +0.77 V):

$VO_2^+ \rightarrow VO^{2+}$: 1.00 − 0.77 = **+0.23 V** ✓; $VO^{2+} \rightarrow V^{3+}$: 0.34 − 0.77 = **−0.43 V** ✗. Fe^{2+} only takes vanadium to +4.

Iodide ($I_2 + 2e^- \rightleftharpoons 2I^-$, +0.54 V): $VO_2^+ \rightarrow VO^{2+}$ is +0.46 V ✓; $VO^{2+} \rightarrow V^{3+}$ is −0.20 V ✗. Also stops at +4.

Iron metal ($Fe^{2+} + 2e^- \rightleftharpoons Fe$, −0.44 V): +1.44, +0.78 and +0.18 V, so iron takes vanadium down to **V^{2+}**. Tin (Sn^{2+}/Sn, −0.14 V) gives +1.14 and +0.48 V but −0.12 V for $V^{3+} \rightarrow V^{2+}$, so tin stops at **$V^{3+}$**.

Oxidation back up

- Air ($O_2 + 4H^+ + 4e^- \rightleftharpoons 2H_2O$, +1.23 V) readily oxidises V^{2+} and V^{3+} up to VO^{2+}. Going on to VO_2^+ has E°cell = 1.23 − 1.00 = +0.23 V: feasible on paper but very slow, so blue VO^{2+} solutions are stable in air.
- Acidified **MnO_4^-** (+1.51 V) oxidises every lower state all the way to **VO_2^+**. This is used to titrate vanadium (16.15).

MARK-SCHEME WORDING

- Iron and tin (6-mark, WCH15 Jan 2023): "both reduce V(V) to V(IV)" (Fe +1.44, Sn +1.14 V); "both reduce V(IV) to V(III)" (+0.78, +0.48 V); "only iron reduces V(III) to V(II)" (+0.18 V; tin −0.12 V). Equations for each step.
- "VO^{2+} is not oxidised any further as E°cell is −0.2 V / negative" (WCH15 Oct 2020).

Often asked as: which reagent reduces vanadium to which state (with equations and E°cell values); 6-mark comparing two reducing agents; balanced equation for MnO_4^- with a vanadium ion.

COMMON MISTAKE

Calculating every E°cell but writing no equations, or the reverse (WCH15 Jan 2023 examiner report). Not saying the reduction **stops** at a given state and why.

Pairing the reduction of VO_2^+ with the wrong oxidation half-equation (WCH15 Oct 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Potassium vanadium alum $KV(SO_4)_2 \cdot 12H_2O$ deduced from % composition (WCH15 Jan 2023). Ferrovanadium analysis by back titration (WCH15 Oct 2022). Equation for MnO_4^- with V^{3+} taking it to V(+5): $2MnO_4^- + 5V^{3+} + 2H_2O \rightarrow 2Mn^{2+} + 5VO_2^+ + 4H^+$.

SPEC 17.20 Reducing and making dichromate(VI)

WHAT THIS SPEC POINT ASKS

Understand, using E° values, that (i) zinc in acid reduces $Cr_2O_7^{2-}$ to Cr^{3+} and then Cr^{2+}, and (ii) Cr^{3+} is oxidised to dichromate(VI) by hydrogen peroxide in alkali, followed by acidification.

Half-equation	E° / V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33
$H_2O_2 + 2e^- \rightleftharpoons 2OH^-$ (alkaline)	+0.88
$CrO_4^{2-} + 4H_2O + 3e^- \rightleftharpoons Cr(OH)_3 + 5OH^-$ (alkaline)	−0.13
$Cr^{3+} + e^- \rightleftharpoons Cr^{2+}$	−0.41
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	−0.76

(i) Zinc reduces dichromate(VI): orange → green → blue

WORKED EXAMPLE

Worked example: zinc and acidified dichromate

Cr(+6) → Cr(+3): E°cell = 1.33 − (−0.76) = **+2.09 V** ✓

$Cr_2O_7^{2-} + 14H^+ + 3Zn \rightarrow 2Cr^{3+} + 7H_2O + 3Zn^{2+}$ (orange → green)

Cr(+3) → Cr(+2): E°cell = −0.41 − (−0.76) = **+0.35 V** ✓

$2Cr^{3+} + Zn \rightarrow 2Cr^{2+} + Zn^{2+}$ (green → blue)

Compare Fe^{2+} (+0.77 V): it reduces $Cr_2O_7^{2-}$ to Cr^{3+} (1.33 − 0.77 = +0.56 V) but **not** to Cr^{2+} (−0.41 − 0.77 = −1.18 V).

- Blue Cr^{2+} is **rapidly oxidised by air** back to green Cr^{3+} (O_2/H_2O at +1.23 V gives +1.64 V). So keep air out: a loose cotton-wool plug, or a bung with a valve, lets H_2 escape but stops air entering.

(ii) Making dichromate(VI) from chromium(III)

1. Add excess **NaOH** to Cr^{3+}(aq): the green precipitate dissolves to dark green **$[Cr(OH)_6]^{3-}$**.
2. Add **hydrogen peroxide** and **warm**: the solution turns **yellow** as chromate(VI) forms.

- $E°_{cell}$ = 0.88 − (−0.13) = **+1.01 V** ✓ (feasible in alkali) - $2[Cr(OH)_6]^{3-} + 3H_2O_2 \rightarrow 2CrO_4^{2-} + 2OH^- + 8H_2O$ - or, from the hydroxide: $2Cr(OH)_3 + 3H_2O_2 + 4OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$

1. Boil off excess H_2O_2 (otherwise it would reduce the dichromate in acid), then **acidify**: yellow CrO_4^{2-} turns **orange** $Cr_2O_7^{2-}$ (17.21).

EXAM TIP

Why alkaline conditions? In acid the margin is small (1.77 − 1.33 = +0.44 V), and H_2O_2 can also act as a reducing agent towards dichromate, so the product doesn't build up cleanly. In alkali the Cr(VI)/Cr(III) potential is far less positive (−0.13 V), so H_2O_2 oxidises Cr(III) easily.

MARK-SCHEME WORDING

- "zinc converts dichromate(VI) to chromium(III) with $E°_{cell}$ +2.09 V and then to chromium(II) with $E°_{cell}$ +0.35 V"; "iron(II) converts dichromate(VI) to chromium(III) (+0.56 V) but not to chromium(II) as … −1.18 V"; "acidified solution" for the conditions (WCH15 Jun 2024).
- Colours: "$Cr_2O_7^{2-}$ orange and Cr^{3+} green"; DO NOT ACCEPT blue for Cr^{3+} (WCH15 Jun 2022).
- Cr^{2+} cell drift: "Cr^{2+} is (rapidly) oxidised (by oxygen in the air)" (WCH15 Jun 2024).

Often asked as: which reagents reduce dichromate to which state (equations + $E°_{cell}$); oxidising Cr(III) with H_2O_2; colours at each stage.

COMMON MISTAKE

Leaving H_2O_2 out of the equation; not cancelling H^+, H_2O or e^- (WCH15 Jan 2025 examiner report).

Comparing E° values in words instead of calculating $E°_{cell}$; using the wrong half-equation from a numbered list (WCH15 Jun 2024 examiner report).

Calling Cr^{3+} "blue" or "blue-green" (WCH15 Jun 2022 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The mark schemes are not consistent on the H_2O_2 equation. One accepted the acid form $2Cr^{3+} + 3H_2O_2 + H_2O \rightarrow Cr_2O_7^{2-} + 8H^+$ (WCH15 Jan 2025). An MCQ used the alkaline form $2Cr(OH)_3 + 3H_2O_2 + 4KOH \rightarrow 2K_2CrO_4 + 8H_2O$ (WCH15 Jun 2023). Learn the alkaline route the spec describes.

Acidified dichromate oxidising ethanol: $3C_2H_5OH + Cr_2O_7^{2-} + 8H^+ \rightarrow 2Cr^{3+} + 3CH_3CHO + 7H_2O$ (WCH15 Jan 2021).

Writing a half-equation in alkaline conditions

1. Start with the change: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-}$.
2. Balance oxygen with water and hydrogen with H^+, as if in acid: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+$.
3. Balance charge with electrons: $[Cr(OH)_6]^{3-} \rightarrow CrO_4^{2-} + 2H_2O + 2H^+ + 3e^-$.
4. Add OH^- to both sides to turn each H^+ into water: $[Cr(OH)_6]^{3-} + 2OH^- \rightarrow CrO_4^{2-} + 4H_2O + 3e^-$.

SPEC 17.21 The dichromate–chromate equilibrium

WHAT THIS SPEC POINT ASKS

Know that dichromate(VI) converts to chromate(VI) through the equilibrium $Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$.

$Cr_2O_7^{2-}(aq) + H_2O(l) \rightleftharpoons 2CrO_4^{2-}(aq) + 2H^+(aq)$ orange ⇌ yellow

- **Add alkali** (OH^-): it removes H^+ ($H^+ + OH^- \rightarrow H_2O$), so the equilibrium shifts **right** → **yellow** chromate(VI).
- **Add acid** (H^+): the equilibrium shifts **left** → **orange** dichromate(VI).
- Chromium is **+6 on both sides**: this is an **acid–base** equilibrium, **not redox**.

MARK-SCHEME WORDING

- "the equilibrium shifts to the chromate(VI) ions because hydrogen ions are removed (as they react with hydroxide)" (WCH15 Oct 2025).
- "not redox because the oxidation number of chromium has not changed: +6 in both" (WCH15 Jan 2025).

Often asked as: write the equation (only 25% got it right, WCH15 Jan 2021); explain the colour change on adding alkali; is it redox?

COMMON MISTAKE

Saying adding alkali "reduces" dichromate. No oxidation number changes.

SPEC 17.22 Reactions of aqueous ions with NaOH and NH_3

WHAT THIS SPEC POINT ASKS

Record observations and write equations for the reactions of Cr^{3+}, Mn^{2+}, Fe^{2+}, Fe^{3+}, Co^{2+}, Ni^{2+}, Cu^{2+} and Zn^{2+}(aq) with aqueous sodium hydroxide and aqueous ammonia, including in excess.

With sodium hydroxide

Ion (colour)	A few drops of NaOH(aq)	Excess NaOH(aq)
Cr^{3+} (green)	green (grey-green) ppt, $[Cr(OH)_3(H_2O)_3]$	ppt **dissolves**: dark green solution, $[Cr(OH)_6]^{3-}$
Mn^{2+} (very pale pink)	off-white / pale brown ppt, $[Mn(OH)_2(H_2O)_4]$; darkens to brown in air	no change
Fe^{2+} (pale green)	green ppt, $[Fe(OH)_2(H_2O)_4]$; turns orange-brown on standing in air	no change
Fe^{3+} (yellow-brown)	red-brown / orange-brown ppt, $[Fe(OH)_3(H_2O)_3]$	no change
Co^{2+} (pink)	blue ppt, $[Co(OH)_2(H_2O)_4]$; turns pink/beige on standing	no change
Ni^{2+} (green)	green ppt, $[Ni(OH)_2(H_2O)_4]$	no change
Cu^{2+} (blue)	(pale) blue ppt, $[Cu(OH)_2(H_2O)_4]$	no change
Zn^{2+} (colourless)	white ppt, $[Zn(OH)_2(H_2O)_4]$	ppt **dissolves**: colourless solution, $[Zn(OH)_4]^{2-}$

Only the **chromium(III)** and **zinc** hydroxides dissolve in excess NaOH: they are **amphoteric**.

With aqueous ammonia

Ion	A few drops of NH_3(aq)	Excess NH_3(aq)
Cr^{3+}	green ppt	ppt dissolves (slowly, best with conc NH_3): **purple / violet** solution, $[Cr(NH_3)_6]^{3+}$
Mn^{2+}	off-white / pale brown ppt (darkens in air)	no change

Ion	A few drops of $NH_3(aq)$	Excess $NH_3(aq)$
Fe^{2+}	green ppt (turns brown in air)	no change
Fe^{3+}	red-brown ppt	no change
Co^{2+}	blue ppt	ppt dissolves: **straw / yellow-brown** solution, $[Co(NH_3)_6]^{2+}$; darkens to brown in air
Ni^{2+}	green ppt	ppt dissolves: **blue** (lavender-blue) solution, $[Ni(NH_3)_6]^{2+}$
Cu^{2+}	pale blue ppt	ppt dissolves: **deep blue** solution, $[Cu(NH_3)_4(H_2O)_2]^{2+}$
Zn^{2+}	white ppt	ppt dissolves: **colourless** solution, $[Zn(NH_3)_4]^{2+}$

With a few drops, ammonia acts as a **base** and gives the **same precipitates** as NaOH. In excess, it acts as a **ligand** for Cr, Co, Ni, Cu and Zn.

Air oxidation on standing

- $Fe(OH)_2$ (green) → $Fe(OH)_3$ (brown): $4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3$. The brown appears first at the **surface**, where the air is.
- $Mn(OH)_2$ similarly darkens to brown as Mn(II) is oxidised to a higher state.

MARK-SCHEME WORDING

- Cobalt (6-mark, WCH15 Jan 2022): pink solution; with NaOH "blue precipitate / pink on standing"; with NH_3 "straw / yellow / yellow-brown / brown solution (ligand exchange)"; with conc HCl "(dark) blue solution". "The initial colour must be mentioned once."
- Nickel (6-mark, WCH15 Jun 2022): "both alkalis initially give a green precipitate"; "deprotonation"; "with excess ammonia a blue solution, no change with NaOH"; "ligand exchange". DO NOT ACCEPT blue-green or blue-purple.
- Zinc (WCH15 Jun 2022): "white precipitate"; "the precipitate dissolves" in excess NaOH "due to formation of $Zn(OH)_4^{2-}$".
- Iron: "$Fe(OH)_2$ is green (initially)" and "turns brown on standing" (WCH15 Jun 2021).

Often asked as: 6-mark "describe what you would see and write equations" for one metal with both reagents; identify an ion from observations; MCQ on colour sequences.

COMMON MISTAKE

Leaving out the **starting colour** of the solution, or the word "precipitate" (WCH15 Jan 2022 examiner report). The most common score on the cobalt 6-mark was 0.

Copying the copper ammine formula for cobalt or nickel: Cu gives $[Cu(NH_3)_4(H_2O)_2]^{2+}$, but Co and Ni give $[M(NH_3)_6]^{2+}$ (WCH15 Jan 2022 examiner report).

Not recognising a red-brown precipitate as iron(III) hydroxide (WCH15 Oct 2020 examiner report).

SPEC 17.23 Equations: deprotonation, amphoteric behaviour and ligand exchange

WHAT THIS SPEC POINT ASKS

Write ionic equations showing amphoteric behaviour, deprotonation and ligand exchange for the reactions in 17.22.

1. Deprotonation (a few drops of OH^- or NH_3)

Hexaaqua ions are **weak acids**: the metal ion pulls electron density from the O–H bonds of its water ligands, so a ligand can lose H^+. A base removes **one H^+ per positive charge**, until the complex is **neutral** and precipitates.

- 2+ ions: **$[M(H_2O)_6]^{2+}(aq) + 2OH^-(aq) \rightarrow [M(OH)_2(H_2O)_4](s) + 2H_2O(l)$**
- 3+ ions: **$[M(H_2O)_6]^{3+}(aq) + 3OH^-(aq) \rightarrow [M(OH)_3(H_2O)_3](s) + 3H_2O(l)$**
- With ammonia as the base: **$[M(H_2O)_6]^{2+}(aq) + 2NH_3(aq) \rightarrow [M(OH)_2(H_2O)_4](s) + 2NH_4^+(aq)$**

2. Amphoteric behaviour (excess OH^- with Cr and Zn)

An **amphoteric** hydroxide reacts with **both acids and bases**.

	As a base (with acid)	As an acid (with excess OH^-)
Cr	$[Cr(OH)_3(H_2O)_3] + 3H^+ \rightarrow [Cr(H_2O)_6]^{3+}$	$[Cr(OH)_3(H_2O)_3] + 3OH^- \rightarrow [Cr(OH)_6]^{3-} + 3H_2O$
Zn	$[Zn(OH)_2(H_2O)_4] + 2H^+ \rightarrow [Zn(H_2O)_6]^{2+}$	$[Zn(OH)_2(H_2O)_4] + 2OH^- \rightarrow [Zn(OH)_4]^{2-} + 4H_2O$

$[Zn(OH)_4]^{2-}$ is tetrahedral, which is why four water ligands are released.

3. Ligand exchange (excess NH_3)

NH_3 replaces water (and hydroxide) ligands. Write it from the precipitate or directly from the aqua ion:

- **Cr**: $[Cr(OH)_3(H_2O)_3] + 6NH_3 \rightarrow [Cr(NH_3)_6]^{3+} + 3OH^- + 3H_2O$
- **Co**: $[Co(OH)_2(H_2O)_4] + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 2OH^- + 4H_2O$; overall $[Co(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 6H_2O$
- **Ni**: $[Ni(OH)_2(H_2O)_4] + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 2OH^- + 4H_2O$; overall $[Ni(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 6H_2O$

- **Cu**: $[Cu(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2OH^- + 2H_2O$; overall $[Cu(H_2O)_6]^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 4H_2O$
- **Zn**: $[Zn(OH)_2(H_2O)_4] + 4NH_3 \rightarrow [Zn(NH_3)_4]^{2+} + 2OH^- + 4H_2O$

EXAM TIP

Check every equation three ways: **atoms** (count O and H carefully), **charge**, and **brackets**. The number of OH^- in the precipitate always equals the charge on the metal ion.

MARK-SCHEME WORDING

- "$[Ni(H_2O)_6]^{2+} + 2OH^- \rightarrow [Ni(H_2O)_4(OH)_2] + 2H_2O$"; "these are deprotonation reactions"; "$[Ni(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Ni(NH_3)_6]^{2+} + 6H_2O$"; "with excess ammonia it is ligand exchange" (WCH15 Jun 2022).
- Reaction type for a little NH_3: "deprotonation"; ammonia "is a base" (WCH15 Jan 2026).
- Square brackets may be omitted; equations written with simple ions ($Ni^{2+} + 2OH^- \rightarrow Ni(OH)_2$) are often accepted.

Often asked as: write the equation for each observation and name the reaction type; MCQ on the equation for an amphoteric hydroxide dissolving in acid.

COMMON MISTAKE

Mixing up **deprotonation** (precipitate forms) and **ligand exchange** (precipitate dissolves in excess NH_3) (WCH15 Jun 2022 examiner report).

Bracket errors like $[Ni(H_2O)_4(OH_2)]$; $[Zn(OH)_6]^{4-}$ instead of **$[Zn(OH)_4]^{2-}$**; NH_4^+ in an equation with OH^- as the base (WCH15 Jun 2022 examiner report).

Writing $H_2O(aq)$: water is (l) (WCH15 Jun 2024 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Predicting an unfamiliar metal's reactions from the pattern (e.g. working out cobalt by analogy with copper, but with six NH_3) (WCH15 Jan 2022).

Sodium carbonate also gives the hydroxide precipitates with 3+ ions (WCH15 Jun 2024).

SPEC 17.24 Ligand exchange and colour changes with copper and cobalt

WHAT THIS SPEC POINT ASKS

Understand the ligand exchange and colour change in forming (i) $[Cu(NH_3)_4(H_2O)_2]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ via $Cu(OH)_2(H_2O)_4$, (ii) $[CuCl_4]^{2-}$ from $[Cu(H_2O)_6]^{2+}$, (iii) $[CoCl_4]^{2-}$ from $[Co(H_2O)_6]^{2+}$.

(i) Copper with ammonia

Stage	Species	Observation	Reaction type
start	$[Cu(H_2O)_6]^{2+}$	blue solution	
a few drops of NH_3	$[Cu(OH)_2(H_2O)_4]$	pale blue precipitate	deprotonation (acid–base)
excess NH_3	$[Cu(NH_3)_4(H_2O)_2]^{2+}$	**deep blue** solution	ligand exchange

- Only **four** of the six water ligands are replaced. The coordination number stays **6** (NH_3 and H_2O are similar in size and both neutral) and the shape stays octahedral.

(ii) Copper with concentrated hydrochloric acid

$[Cu(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CuCl_4]^{2-}(aq) + 6H_2O(l)$ blue → (green) → **yellow**

- Coordination number **6** → **4**; octahedral → **tetrahedral** (Cl^- is large). Charge 2+ → 2−. Copper stays **+2**.
- The green seen part-way is a **mixture** of blue and yellow complexes.
- It is **reversible**: adding water shifts the equilibrium back to the blue aqua ion.

(iii) Cobalt with concentrated hydrochloric acid

$[Co(H_2O)_6]^{2+}(aq) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq) + 6H_2O(l)$ pink → blue

- Again 6 → 4, octahedral → tetrahedral, cobalt stays +2. Adding water turns it pink again.
- The forward reaction is endothermic, so warming the mixture makes it bluer. (This is the chemistry of cobalt chloride paper: blue when dry, pink with water.)

MARK-SCHEME WORDING

- Copper 6-mark (WCH15 Jun 2025): "tetrahedral ($[CuCl_4]^{2-}$) to octahedral ($[Cu(H_2O)_6]^{2+}$ / $[Cu(NH_3)_4(H_2O)_2]^{2+}$)"; "the Cl^- ion is larger, so more water ligands can fit"; "copper(II) hydroxide

precipitates as the solution becomes alkaline"; "yellow to blue solution, or precipitate to deep blue, is ligand exchange". DO NOT ACCEPT "blue solution to precipitate" as ligand exchange.

- "$[Co(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CoCl_4]^{2-} + 6H_2O$". IGNORE state symbols (WCH15 Oct 2022).
- DO NOT ACCEPT $[Cu(NH_3)_6]^{2+}$ for the deep blue complex (WCH15 Jan 2021).

Often asked as: 6-mark tracing colour and shape changes around copper(II); equations for the chloro complexes; MCQ on the colour of $[CuCl_4]^{2-}$.

COMMON MISTAKE

Calling the precipitate step "ligand exchange" (it is deprotonation). Writing $[Cu(NH_3)_6]^{2+}$.

Saying copper is reduced when the solution goes yellow: the oxidation state is unchanged.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

The order of exchange follows how readily each ligand donates its lone pair: **Cl^- < H_2O < NH_3**, so water replaces chloride and ammonia replaces water (WCH15 Jun 2025).

Cu(I) ammine complexes, colourless $[Cu(NH_3)_2]^+$, oxidised by air to the deep blue Cu(II) complex (WCH15 Jan 2021).

SPEC 17.25 The chelate effect: entropy and stability

WHAT THIS SPEC POINT ASKS

Understand, in terms of a positive ΔS(system), why replacing a monodentate ligand by a bidentate or hexadentate ligand gives a more stable complex.

Count the particles

Reaction	Particles before → after
$[Cu(H_2O)_6]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 6H_2O$	4 → **7**
$[Cu(H_2O)_6]^{2+} + EDTA^{4-} \rightarrow [Cu(EDTA)]^{2-} + 6H_2O$	2 → **7**
$[Cu(NH_3)_4(H_2O)_2]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 4NH_3 + 2H_2O$	4 → **7**

- More particles in solution means more ways of arranging them, so **ΔS(system) is positive**.

- **ΔH is about zero**: the same number and the same kind of dative bonds are broken and made (six M–O or M–N bonds each side).
- So ΔS(surroundings) = −ΔH/T ≈ 0, and **ΔS(total) is positive**. Equivalently **ΔG = ΔH − TΔS is negative**. The reaction is feasible with a **large K**, so the chelate complex is **more stable** and the reverse reaction hardly happens.

WORKED EXAMPLE

Worked example: EDTA and a hexaaqua ion

$[Ni(H_2O)_6]^{2+}(aq) + EDTA^{4-}(aq) \rightarrow [Ni(EDTA)]^{2-}(aq) + 6H_2O(l)$

2 particles on the left, 7 on the right → ΔS(system) is large and positive.

ΔH ≈ 0, because six dative bonds to O are replaced by six dative bonds to N and O.

So ΔS(total) > 0 and ΔG < 0: the EDTA complex forms and is very stable.

MARK-SCHEME WORDING

- "there are **more particles / moles** on the right (of the equation)" and "so **ΔS(system) increases / is positive** (and the reaction is thermodynamically feasible)". Marks are independent (WCH15 Jun 2024; Oct 2023).
- "Allow species or ions for particles". DO NOT ACCEPT "molecules", "atoms" or "compounds" (WCH15 Oct 2023; Jun 2022).
- IGNORE "entropy is positive" (entropy is always positive; it is the **change** that is positive).

Often asked as: write the equation for $EDTA^{4-}$ or en replacing water and explain the stability (2–4 marks); count particles in an unfamiliar ligand exchange.

COMMON MISTAKE

"More molecules on the right" (the products include ions). "Entropy is positive" instead of "the entropy **change** of the system is positive" (WCH15 Jun 2022; Oct 2023 examiner reports).

Talking about bond strength instead of entropy (WCH15 Oct 2022 examiner report). Putting Na^+ inside the complex (WCH15 Jun 2024 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Counting particles when some species are polyatomic anions, e.g. 5 on the left and 7 on the right (WCH15 Jun 2022).

SPEC 17.26 Transition metals as catalysts

WHAT THIS SPEC POINT ASKS

Know that transition metals and their compounds act as heterogeneous and homogeneous catalysts.

- A catalyst provides an **alternative route with a lower activation energy**. It doesn't change ΔH or the position of equilibrium (K), and is unchanged at the end.
- Transition metals are good catalysts because:

- they have **variable oxidation states**, so they can accept and donate electrons, then return to the original state (homogeneous catalysis and V_2O_5); - their **d orbitals** (and partly filled d subshells) let reactant molecules bond to the **surface** (heterogeneous catalysis).

Catalyst	Type	Process
Fe	heterogeneous	Haber process: $N_2 + 3H_2 \rightleftharpoons 2NH_3$
Ni	heterogeneous	hydrogenation of alkenes (e.g. making margarine)
V_2O_5	heterogeneous	Contact process: $2SO_2 + O_2 \rightleftharpoons 2SO_3$
Pt / Pd / Rh	heterogeneous	catalytic converters
Fe^{2+} / Fe^{3+}	homogeneous	$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$
Mn^{2+}	homogeneous (autocatalyst)	$MnO_4^- + C_2O_4^{2-}$

MARK-SCHEME WORDING

Compare and contrast (6-mark, WCH15 Oct 2020): both "provide an alternative route / mechanism with a lower activation energy"; "heterogeneous is in a different phase from the reactants, homogeneous in the same phase"; examples (Fe in the Haber process, Ni in hydrogenation, Pt in a catalytic converter; Fe^{2+}/Fe^{3+} with I^- and $S_2O_8^{2-}$); heterogeneous "adsorbed … bonds weakened … reaction … desorbed"; homogeneous "the transition metal ion is oxidised / reduced to a different oxidation state, then changes back".

Often asked as: 6-mark compare heterogeneous and homogeneous catalysis; MCQ on catalyst types.

COMMON MISTAKE

Writing about the two types separately in a "compare" question. Pair them ("**both** lower Ea, **but** …") (WCH15 Oct 2020 examiner report).

Mixing up the Haber (Fe) and Contact (V_2O_5) catalysts.

SPEC 17.27 Heterogeneous catalysts

WHAT THIS SPEC POINT ASKS

Know that a heterogeneous catalyst is in a different phase from the reactants and that the reaction happens on its surface.

- **Heterogeneous**: the catalyst is in a **different phase** (usually a solid with gas or liquid reactants).
- The reaction happens at **active sites** on the **surface**:

1. **Adsorption**: reactant molecules form weak bonds to the surface. 2. **Reaction**: bonds within the reactants are **weakened**, and the reactants are held close together in a good orientation, so they react. 3. **Desorption**: the products leave the surface, freeing the sites.

- **Adsorption strength** must be just right. Too strong (e.g. W) and the products don't leave; too weak (e.g. Ag) and the reactants don't stick. Ni and Pt are in between.
- To get a large **surface area** cheaply, the metal is spread thinly on a **support** (e.g. a ceramic honeycomb).
- **Poisoning**: impurities adsorb strongly and block active sites (lead on catalytic converters; sulfur on the iron in the Haber process). Solid catalysts are easy to separate from products.

MARK-SCHEME WORDING

- "adsorption … to the surface"; "results in the weakening of bonds"; "reaction occurs"; "products desorb". DO NOT ACCEPT "absorption" (penalised once) (WCH15 Oct 2025).
- ACCEPT "easy to separate" as an advantage of heterogeneous catalysts (WCH15 Oct 2020).

COMMON MISTAKE

"Absorb" for **adsorb**. Absorb means soak into; adsorb means stick onto the surface.

SPEC 17.28 Vanadium(V) oxide in the Contact process

WHAT THIS SPEC POINT ASKS

Understand, in terms of oxidation number, how V_2O_5 catalyses the Contact process.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ (about 450 °C, 1–2 atm, V_2O_5 catalyst)

V_2O_5 is a solid catalyst for gases (heterogeneous), but it works by **changing oxidation state**:

1. **$SO_2 + V_2O_5 \rightarrow SO_3 + V_2O_4$**: V is reduced, **+5 → +4**; S is oxidised, +4 → +6.
2. **$V_2O_4 + \frac{1}{2}O_2 \rightarrow V_2O_5$**: V is oxidised back, **+4 → +5**.

Adding the two steps gives $SO_2 + \frac{1}{2}O_2 \rightarrow SO_3$: the V_2O_5 is regenerated.

MARK-SCHEME WORDING

- "$V_2O_5 + SO_2 \rightarrow V_2O_4 + SO_3$; $V_2O_4 + \frac{1}{2}O_2 \rightarrow V_2O_5$". ACCEPT any balanced pair showing a lower oxide formed by SO_2 then reoxidised by O_2. Two equations in the wrong order score 1 (WCH15 Oct 2022).
- DO NOT ACCEPT V_2O_5 described as a **homogeneous** catalyst (WCH15 Jan 2021).

Often asked as: two equations with oxidation states; MCQ on vanadium's oxidation state during the cycle.

COMMON MISTAKE

Calling V_2O_5 homogeneous because it changes oxidation state. It is a solid catalysing gases: **heterogeneous** (WCH15 Jan 2021 examiner report).

SPEC 17.29 Catalytic converters

WHAT THIS SPEC POINT ASKS

Understand how a catalytic converter cuts CO and NO emissions: CO and NO adsorb on the catalyst surface, bonds weaken and they react; then CO_2 and N_2 desorb.

- **Where the pollutants come from**: CO from incomplete combustion of fuel; NO from N_2 and O_2 in air reacting at the high temperature of the engine.
- **The catalyst**: platinum, palladium and rhodium, coated thinly on a **ceramic honeycomb** (large surface area, little expensive metal).

$2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)$

1. **Adsorption**: CO and NO molecules form bonds to the metal surface.
2. **Bonds weaken** (e.g. the N–O bond), and the adsorbed species react.
3. **Desorption**: CO_2 and N_2 leave the surface, freeing the active sites.

- Unburnt hydrocarbons are also oxidised to CO_2 and H_2O.
- Lead (from old leaded petrol) **poisons** the catalyst, which is why converters need unleaded fuel.

MARK-SCHEME WORDING

- Jan 2021 6-mark (V_2O_5 v Pt in a converter): both heterogeneous; adsorption of reactants (both); weakening of bonds (converter); desorption of products (both). DO NOT ACCEPT "chemisorption" or "bond strongly" (WCH15 Jan 2021).
- CO on Pt (WCH15 Oct 2025): "adsorption of CO to the surface"; "weakening of bonds"; "reaction / oxidation occurs"; "to form CO_2"; "which then desorbs". A reference to the same state or phase loses a mark.

Often asked as: describe how the converter removes CO and NO (3–6 marks); MCQ: "CO and NO are **ad**sorbed".

COMMON MISTAKE

Missing out **desorption** of products (WCH15 Jan 2021 examiner report). Writing "absorbed".

SPEC 17.30 Homogeneous catalysts

WHAT THIS SPEC POINT ASKS

Know that a homogeneous catalyst is in the same phase as the reactants, and that the catalysed reaction goes via an intermediate species.

- **Homogeneous**: catalyst and reactants are in the **same phase**, usually all in aqueous solution.
- The catalyst **reacts** with one reactant to form an **intermediate**, which then reacts to give the products and **regenerate** the catalyst.
- Transition-metal ions are good at this because they switch easily between **oxidation states** (Fe^{2+}/Fe^{3+}, Mn^{2+}/Mn^{3+}).
- The energy profile shows **two humps** (one per step), each lower than the single uncatalysed hump. ΔH is the same.

MARK-SCHEME WORDING

- "same phase"; ACCEPT "same state"; "all species are in aqueous solution" (WCH15 Jun 2021).
- "the transition metal ion is oxidised / reduced to a different oxidation state then changes back" (WCH15 Oct 2020).

SPEC 17.31 Fe^{2+} catalysing the reaction of I^- with $S_2O_8^{2-}$

WHAT THIS SPEC POINT ASKS

Understand how Fe^{2+} ions catalyse the reaction between iodide and peroxodisulfate ions.

$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$

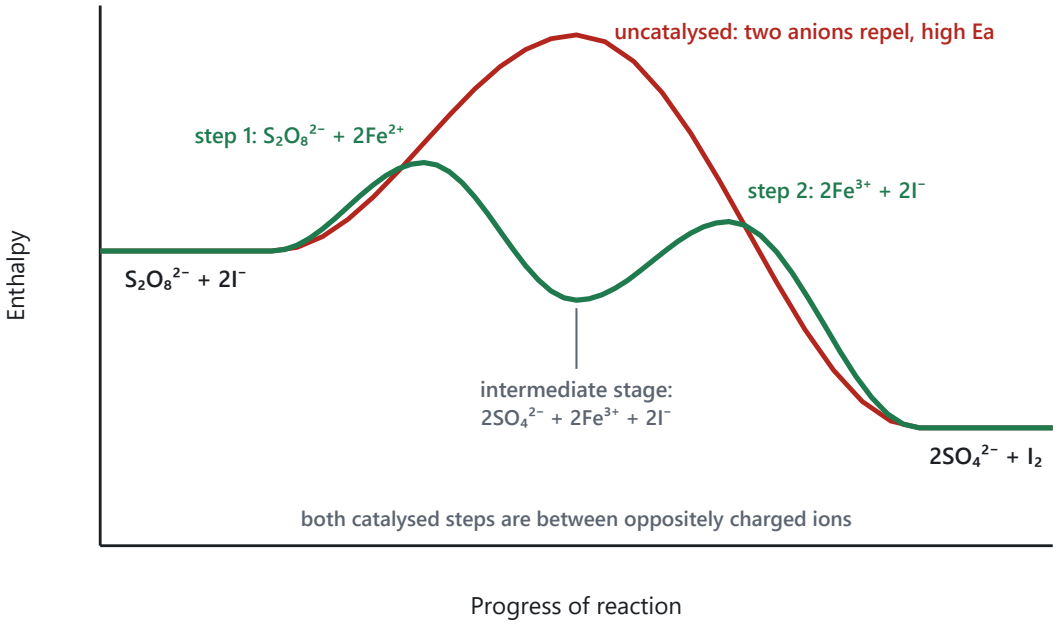
Why it is slow without a catalyst

E°cell = 2.01 − 0.54 = **+1.47 V**, so it is very feasible. But both reactants are **negative ions**: they **repel**, so the activation energy is high and the reaction is slow (16.12).

The catalysed route

Step 1: $S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$ E°cell = 2.01 − 0.77 = **+1.24 V Step 2**: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$ E°cell = 0.77 − 0.54 = **+0.23 V**

- Each step is between **oppositely charged ions**, which attract, so each has a lower activation energy.
- Fe^{2+} is regenerated in step 2. **Fe^{3+} works equally well**: it just starts with step 2.
- The method works because **E°(Fe^{3+}/Fe^{2+}) = +0.77 V lies between** E°(I_2/I^-) = +0.54 V and E°($S_2O_8^{2-}/SO_4^{2-}$) = +2.01 V, so both steps have positive E°cell.



The catalysed route has two lower humps. The same overall ΔH.

MARK-SCHEME WORDING

- Equations: "$S_2O_8^{2-} + 2Fe^{2+} \rightarrow 2SO_4^{2-} + 2Fe^{3+}$" and "$2I^- + 2Fe^{3+} \rightarrow I_2 + 2Fe^{2+}$" (either order) (WCH15 Jun 2021).
- Why catalysed: "the uncatalysed reaction involves two negatively charged ions reacting, and the catalysed steps involve oppositely charged ions" (WCH15 Jun 2021); "reaction between two negative ions is slow due to repulsion" (WCH15 Oct 2023). DO NOT ACCEPT "negative molecules".
- "addition of either ion results in a reaction that forms the other ion, so either can act as the catalyst" (WCH15 Jan 2026).

Often asked as: two equations for the catalysed route (2 marks); why the uncatalysed reaction is slow; use E° values to show both steps are feasible; why Fe^{3+} also catalyses.

COMMON MISTAKE

Saying Fe^{2+} catalyses iodide with **thiosulfate** (it is **peroxodisulfate**, $S_2O_8^{2-}$) (WCH15 Oct 2020 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Calculating E°($S_2O_8^{2-}/SO_4^{2-}$) = +2.01 V from a cell measurement, then showing the uncatalysed reaction is feasible but slow (WCH15 Jan 2026). Choosing another possible catalyst: any couple with E° between +0.54 and +2.01 V could in principle work.

Iron ions catalysing the reaction between peroxodisulfate and iodide

1. Overall: $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$. Uncatalysed it is slow, because two negative ions repel each other, so collisions with enough energy are rare.
2. Step 1: $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$
3. Step 2: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$. The iron is back as Fe^{2+}, and Fe^{3+} works equally well because the steps can happen in either order.

4 Both steps are between oppositely charged ions, so they are faster. This works because $E^{\ominus}(Fe^{3+}/Fe^{2+})$ = +0.77 V lies **between** $E^{\ominus}(I_2/I^-)$ = +0.54 V and $E^{\ominus}(S_2O_8^{2-}/SO_4^{2-})$ = +2.01 V, so both steps are feasible.

Electrode potentials show that a catalyst could work, but not that it will speed the reaction up: they say nothing about rates.

SPEC 17.32 Autocatalysis by Mn^{2+}

WHAT THIS SPEC POINT ASKS

Know that Mn^{2+} ions autocatalyse the reaction between MnO_4^- and $C_2O_4^{2-}$ ions.

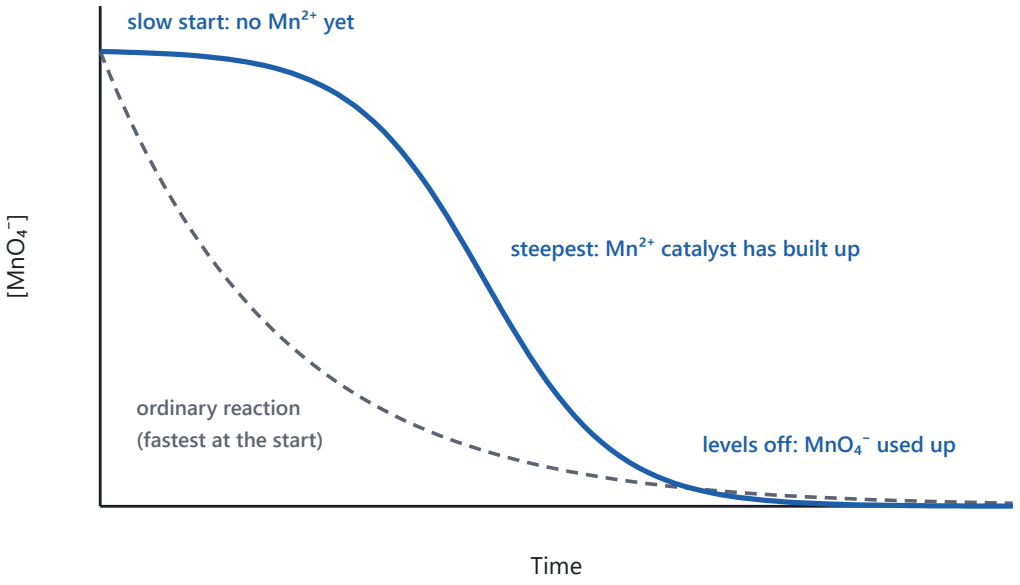
$2MnO_4^-(aq) + 5C_2O_4^{2-}(aq) + 16H^+(aq) \rightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 8H_2O(l)$

- **Autocatalysis**: a **product** of the reaction **catalyses** the reaction. Here the product **Mn^{2+}** is the catalyst.
- **Start**: no Mn^{2+} yet, and both reactants are **anions** that repel, so the reaction is very slow. In a titration the first drops of purple MnO_4^- take a while to fade, even in the warm (≈ 60 °C) flask.
- **Middle**: as Mn^{2+} builds up, the rate **increases**. The purple colour now vanishes instantly.
- **End**: the rate falls as the reactants run out.

The catalytic cycle

1. $MnO_4^- + 4Mn^{2+} + 8H^+ \rightarrow 5Mn^{3+} + 4H_2O$ (Mn^{2+} is oxidised to Mn^{3+})
2. $2Mn^{3+} + C_2O_4^{2-} \rightarrow 2Mn^{2+} + 2CO_2$ (Mn^{3+} oxidises ethanedioate and is reduced back to Mn^{2+})

Each step is between oppositely charged ions.



An autocatalysed reaction gives an S-shaped (sigmoid) concentration–time curve. Follow it with a colorimeter, since MnO_4^- is the only coloured species.

MARK-SCHEME WORDING

- "this reaction is (auto)catalysed by the Mn^{2+} ions formed"; "experiment 1 starts slowly because there is no Mn^{2+} / catalyst present initially"; "experiment 2 is fastest at the start as Mn^{2+} is already present" (WCH15 Jan 2023).
- Rate–time sketch: "(very) slow initial rate which increases as the catalyst / Mn^{2+} forms", "then decreases (and levels off) as the reactant concentration decreases". DO NOT ACCEPT "when all the reactants are used up the reaction starts to slow down" (WCH15 Oct 2025).
- Step equation: "$10Mn^{3+} + 5C_2O_4^{2-} \rightarrow 10Mn^{2+} + 10CO_2$" (any multiple) (WCH15 Oct 2025).

Often asked as: interpret concentration–time graphs with and without added Mn^{2+} (3 marks); sketch the rate–time curve; write the step equations.

COMMON MISTAKE

A stock definition of a catalyst with no reference to the graphs. Saying Mn^{2+} catalyses only one experiment. Treating dissolved $MnSO_4$ as a heterogeneous catalyst with a "surface area" (WCH15 Jan 2023 examiner report).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Rate-determining steps and orders in an unfamiliar catalytic cycle (rhodium catalysis in the Cativa process, WCH15 Jan 2026).

SPEC 17.33 Core Practical 14: preparing a transition-metal complex

WHAT THIS SPEC POINT ASKS

Prepare a transition-metal complex.

Aim and principle

The usual preparation is **tetraamminecopper(II) sulfate**, made by **ligand exchange** of water for ammonia:

$[Cu(H_2O)_6]^{2+}(aq) + 4NH_3(aq) \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}(aq) + 4H_2O(l)$ $CuSO_4 \cdot 5H_2O + 4NH_3 \rightarrow [Cu(NH_3)_4]SO_4 \cdot H_2O + 4H_2O$

- The first drops of ammonia give a **pale blue precipitate** (deprotonation); excess gives a **deep blue solution** (ligand exchange).
- The complex is much **less soluble in ethanol** than in water, so adding ethanol makes it crystallise out.

CORE PRACTICAL

Core practical 14: method in brief

1. Weigh about 2.5 g of $CuSO_4 \cdot 5H_2O$ by difference and dissolve it in the **minimum** volume of distilled water (a few cm^3) in a small beaker.
2. **In a fume cupboard**, add **concentrated ammonia** solution (about 5 cm^3) with stirring: a pale blue precipitate, then a deep blue solution.
3. Slowly add about 8 cm^3 of **ethanol** and cool in an ice bath until deep blue crystals form.
4. Filter by **suction (Buchner) filtration**. **Wash** the crystals with a little **ice-cold ethanol**.
5. Dry between filter papers or in a desiccator. Weigh, and calculate the percentage yield.

Why each step

Step	Reason
minimum water	more of the product crystallises; less is lost in the filtrate
fume cupboard	concentrated NH_3 releases toxic, corrosive ammonia gas
measuring cylinders are fine	volumes are approximate and NH_3 is in excess

Step	Reason
add ethanol	lowers the solubility of the complex
cold ethanol wash	removes soluble impurities without dissolving the product; ethanol evaporates quickly so the crystals dry fast

Safety

Concentrated ammonia: corrosive, irritating vapour (gloves, goggles, fume cupboard). Ethanol: highly flammable, no flames. Copper(II) sulfate: harmful and toxic to aquatic life, so don't pour it down the sink.

WORKED EXAMPLE

Worked example: percentage yield

2.40 g of $CuSO_4 \cdot 5H_2O$ (M = 249.6 g mol^{-1}) gives 1.87 g of $[Cu(NH_3)_4]SO_4 \cdot H_2O$ (M = 245.6 g mol^{-1}).

$n(CuSO_4 \cdot 5H_2O) = 2.40 \div 249.6 = 9.615 \times 10^{-3}$ mol = n(complex) (1 : 1)

theoretical mass = $9.615 \times 10^{-3} \times 245.6 = 2.361$ g

% yield = $1.87 \div 2.361 \times 100$ = **79.2%**

A "yield" above 100% means the crystals were still **damp** (or impure).

Errors and improvements

- Low yield: product left in solution (too much water, or the wash solvent not cold); losses on transfer and filtering.
- Impure product: wash properly with cold ethanol; recrystallise.
- Dry to **constant mass** before the final weighing.

MARK-SCHEME WORDING

- "deep / dark blue solution" (not just "blue"); "less soluble in ethanol, so it crystallises"; "wash with cold ethanol to remove soluble impurities, cold so the product doesn't dissolve".
- The method itself has not been examined in WCH15 so far (it is assessed in Unit 6). WCH15 asks for **formula and hydrate calculations** on prepared salts instead: $FeCl_2 \cdot 4H_2O$, $K_2CuCl_4 \cdot 2H_2O$, $KV(SO_4)_2 \cdot 12H_2O$, $KCr(SO_4)_2 \cdot 12H_2O$, Reinecke's salt (WCH15 Oct 2021; Jan 2021; Jan 2023; Jan 2025; Jan 2024).

Often asked as: deduce the formula of a hydrated complex salt from % composition or titration data; observations on adding NH_3.

COMMON MISTAKE

Using atomic numbers instead of relative atomic masses in a formula calculation (WCH15 Oct 2021 examiner report).

Writing the product as $[Cu(NH_3)_6]^{2+}$.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Other complexes that can be prepared and analysed: potassium tris(ethanedioato)ferrate(III), $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$, and the chromium and vanadium alums. The water of crystallisation is found by heating to constant mass or from titration data (WCH15 Jan 2023; Jan 2025).

Quick checks

QUICK CHECK

What is the electron configuration of Fe^{3+}?

- [Ar] $3d^3\ 4s^2$
- [Ar] $3d^5$
- [Ar] $3d^6$
- [Ar] $3d^4\ 4s^1$

Iron is [Ar] $3d^6\ 4s^2$; the two 4s electrons are lost first, then one 3d electron.

QUICK CHECK

What is the coordination number of chromium in $[Cr(C_2O_4)_3]^{3-}$?

- 3
- 4
- 6
- 9

Each ethanedioate is bidentate, forming two dative bonds: 3 × 2 = 6. The coordination number counts bonds, not ligands.

QUICK CHECK

Why is a solution of $[Zn(H_2O)_6]^{2+}$ colourless?

- zinc has no d electrons
- its 3d sub-shell is full, so no d–d electron transition is possible
- water is a weak ligand
- the complex is tetrahedral

Zn^{2+} is $3d^{10}$. With every d orbital full, an electron cannot move to the upper set, so no visible light is absorbed.

QUICK CHECK

Excess aqueous ammonia is added to aqueous copper(II) sulfate. What is seen?

- a blue precipitate that stays
- a white precipitate
- a blue precipitate that dissolves to give a deep blue solution
- a yellow-green solution

A little ammonia deprotonates the aqua ion, giving blue $[Cu(OH)_2(H_2O)_4]$; excess ammonia then exchanges ligands to form deep blue $[Cu(NH_3)_4(H_2O)_2]^{2+}$.

QUICK CHECK

Why are solutions of $[Fe(H_2O)_6]^{3+}$ more acidic than those of $[Fe(H_2O)_6]^{2+}$?

- Fe^{3+} has more water ligands
- Fe^{3+} has a higher charge density, so it weakens the O–H bonds in its water ligands more
- Fe^{3+} is a Brønsted–Lowry base
- Fe^{2+} forms a precipitate with water

Both have six water ligands. The smaller, more highly charged Fe^{3+} ion polarises them more, so they release H^+ more readily.

QUICK CHECK

Which reaction is favoured mainly by a large increase in entropy?

- $[Cu(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CuCl_4]^{2-} + 6H_2O$
- $[Co(H_2O)_6]^{2+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{2+} + 6H_2O$
- **$[Ni(NH_3)_6]^{2+} + 3en \rightarrow [Ni(en)_3]^{2+} + 6NH_3$**
- $2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$

Four particles become seven when bidentate en replaces ammonia: the chelate effect. ΔH is close to zero because Ni–N bonds are broken and made.

QUICK CHECK

Zinc and acid reduce vanadium(V) to vanadium(II), but tin only reduces it to vanadium(III). Why?

- tin is a stronger reducing agent than zinc
- V^{3+} is unstable
- **$E^{\ominus}(Sn^{2+}/Sn)$ is more negative than $E^{\ominus}(VO^{2+}/V^{3+})$ but not more negative than $E^{\ominus}(V^{3+}/V^{2+})$**
- tin forms a complex with V^{3+}

Sn^{2+}/Sn is −0.14 V. It can reduce couples with more positive $E^{\ominus}$ (+1.00 and +0.34 V) but not V^{3+}/V^{2+} at −0.26 V. Zinc, at −0.76 V, can reduce all three.

QUICK CHECK

In heterogeneous catalysis, why is silver a poor catalyst for many reactions?

- it adsorbs reactants too strongly
- it adsorbs reactants too weakly, so few stay on the surface long enough to react
- it is in the same phase as the reactants
- it changes oxidation state too easily

Too weak adsorption (silver) and too strong adsorption (tungsten) both make poor catalysts. Nickel and platinum adsorb with intermediate strength.